

Does Retrieval Augmented Generation Collapse When It Retrieves AI-Generated Documents?

Riddhimaan Senapati¹ Gregory Druck² Fabeha Fatima¹ Ozel Yilmazel¹
Rati Rastogi¹ Chang Zeng¹ Ethan Smith²

¹University of Massachusetts Amherst ²Graphite

{rsenapati, ffatima, oyilmazel, ratirastogi, changzeng}@umass.edu

{greg, ethan}@graphitehq.com

Abstract

AI-generated text now constitutes a substantial and growing fraction of indexed web content, creating a feedback loop in which AI assistants may retrieve and condition on their own prior generations. Prior work shows that recursive *training* on model generations causes **model collapse**: a loss of output diversity and distributional drift. We ask whether a similar collapse emerges at *inference* time, when LLMs retrieve their own generations via RAG without any parameter updates. We measure collapse as the progressive convergence of responses to a narrow set across recursive retrieval rounds. Across three contamination simulations and four open-source LLMs, collapse is rapid and consistent under direct contamination, but partial and model-dependent under competitive retrieval. Collapse can also degrade downstream accuracy on factual multi-hop QA, though the effect is inconsistent across models. We further show that the self-generated documents we introduce are cited far above their share of the context, an effect that is not fully explained by document quality, writing style, AI authorship, or self-preference. Finally, we propose and evaluate mitigation strategies at both the retrieval and generation stages, none of which reliably prevent collapse. As AI-generated content continues to populate the open web, collapse-resistant retrieval design is necessary for trustworthy AI assistants.

1 Introduction

Large Language Models (LLMs) increasingly rely on Retrieval-Augmented Generation (RAG) to ground their outputs in up-to-date web content. AI-generated text now makes up a substantial and growing fraction of the indexed web: a recent analysis of English-language articles sampled from Common Crawl estimates that primarily AI-generated articles now account for roughly half of all articles published online, a level that has plateaued near parity with human-written articles (Paredes et al.,

2026). This creates a feedback loop in which models retrieve and condition on content that is itself a product of prior model generations.

Prior work has demonstrated that *training* on recursively generated AI content leads to *model collapse*, an irreversible drift in a model’s output distribution that degrades diversity and erodes rare or long-tail knowledge (Shumailov et al., 2024; Mohsin et al., 2025). However, most existing studies focus on the training-time setting, where collapse is induced through parameter updates over successive generations of synthetic data. Unlike training collapse, which affects only models re-trained on contaminated corpora, RAG-induced collapse is an imminent risk for any deployed system. Production search summarizers and deep research agents are continuously exposed to AI-generated retrieved content with no mechanism to detect or filter it. Beyond degrading output quality, a self-reinforcing retrieval loop can amplify biases and make rare information progressively harder to find (Kleinberg and Raghavan, 2021).

Wang et al. (2025) show collapse in a network of LLMs sharing a retrieval corpus. Druck and Smith (2026) show RAG collapse in commercial models, but do not experiment with open models, evaluate factual QA, or test mitigations.

In this paper, we investigate whether and how collapse can occur at inference time through RAG (in the absence of any parameter updates) using open models, measure its effect on downstream factual multi-hop question answering (not only subjective queries), and implement and evaluate mitigations at the retrieval and generation stages.¹

To measure RAG collapse, we adopt the controlled recursive-feedback simulation design of Druck and Smith (2026), in which an assistant repeatedly retrieves documents, conditions on

¹Code and data: <https://github.com/rag-collapse/RAG-Collapse>

them to answer, and regenerates those answers as web-like documents that re-enter the retrieval pool. We evaluate four open-source LLMs under three contamination settings on a subjective entity-comparison QA dataset and HotpotQA (Yang et al., 2018), measuring both diversity collapse and downstream task degradation, and test mitigations.

In all four models, responses grow more similar and mention fewer distinct entities as AI-generated references come to dominate the context. We trace this to a preference for answer-derived documents. Self-generated references are disproportionately cited, and this is not fully explained by reference quality, writing style, or AI authorship. Whereas Druck and Smith (2026) attribute this to self-preference, we show collapse persists when references are generated by a different model. None of the mitigations we test reliably prevent collapse.

2 Related Work

Model Collapse in Training Shumailov et al. (2024) show that recursive training on AI-generated data causes irreversible degradation: models lose tail distributions, output diversity collapses, and perplexity increases over generations.

Collapse Risks at Scale Building on Shumailov et al. (2024), Mohsin et al. (2025) argue that under contaminated data, long-tail knowledge becomes noisier, outdated/conflicting facts are harder to verify, and accumulated bias amplifies hallucinations.

Collapse via RAG at Network Level Wang et al. (2025) simulate LLMs sharing a RAG database on one specific topic and show semantic convergence across models over time; collapse emerges at inference without any retraining. We differ in scale and scope: rather than a single shared topic, we evaluate four models across two datasets and three contamination simulations, and measure both diversity collapse and downstream task degradation.

RAG Collapse in Commercial Models Druck and Smith (2026), by a subset of the authors of this paper, show RAG collapse in commercial models, and argue that collapse is accelerated by model self-preference. In this paper, we experiment with open models, factual QA, a paraphrasing intervention, and mitigations.

3 Methods

3.1 Definition of Collapse

RAG collapse is the progressive concentration of the model’s answer distribution across recursive re-

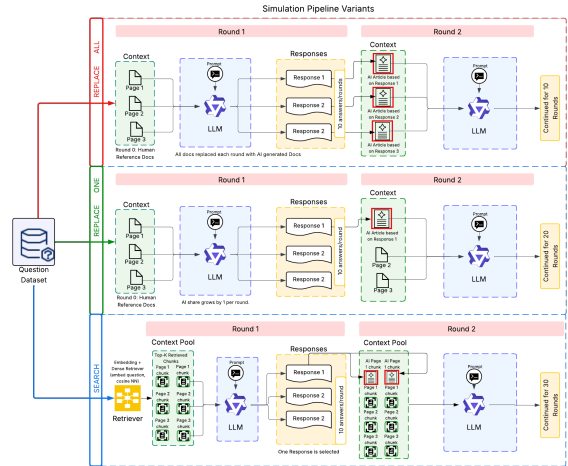


Figure 1: Overview of the three contamination simulations used to model AI-content contamination in RAG systems. From top to bottom, the three rows depict **Replace-All** (the entire context is replaced with generated documents after round 1), **Replace-One** (one reference is replaced per round), and **Search** (self-generated documents are gradually added to that question’s reference pool and compete during retrieval).

trieval rounds, and its drift away from the model’s first-round answers, as AI-generated documents enter the retrieval corpus and influence subsequent generations. We use *collapse* and *RAG collapse* interchangeably for this inference-time phenomenon, *diversity collapse* for its effect on output diversity specifically, and reserve *model collapse* for training-time collapse.

We measure concentration through lexical and entity convergence, and drift through declining accuracy on multi-hop QA (Section 4.2).

3.2 Contamination Simulations

To model how AI-generated content may contaminate retrieval corpora over time, we use the simulation designs of Druck and Smith (2026). Questions are simulated independently. Each simulation iteratively generates answers, rewrites each answer as a web-style document, and updates that question’s retrieval context (Figure 1).

Replace-All replaces the entire context with AI-generated documents after the first round, isolating collapse driven by the retrieval loop itself. This is the inference-time analog of the recursive training studied by Shumailov et al. (2024).

Replace-One replaces one original reference per round with a self-generated document, so synthetic content accumulates gradually.

Search adds one self-generated document per

round to that question’s pool of original references, allowing self-generated and original documents to compete during retrieval.

Document Synthesis. After each round, we convert answers into synthetic web documents using Qwen2.5-7B-Instruct (temperature 0.7), with a prompt that expands each answer into an article while preserving all factual claims (Appendix C). Generating articles rather than feeding answers back directly reflects how AI content appears on the web. We fix the document generator across all experiments to isolate the answer model’s effect.

Retrieval Configuration. For retrieval on the Entity Comparison dataset (the Search simulation, the re-ranker mitigation, and agentic RAG), we embed with all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), split documents into 500-character chunks with 50-character (10%) overlap, and retrieve the top- $k = 10$ chunks by cosine similarity from an in-memory index. For HotpotQA we instead index the corpus with E5-small-v2 (Wang et al., 2024) and FAISS (Johnson et al., 2017), retrieving the top- $k = 10$ passages. We min-max normalize retrieval scores to $[0, 1]$ before any re-ranking to ensure consistent score interpretation across queries. The AI-reference percentage is computed from ground-truth document provenance tags (synthetic vs. original), not from the AI detector; the detector is used only in the re-ranker mitigation.

3.3 Agentic RAG

RAG systems are increasingly agentic, with the model deciding when to retrieve and what queries to issue. If collapse is partly driven by the model repeatedly surfacing similar documents and narrowing its sources across rounds, then letting it actively explore, and separating retrieval from generation, may preserve context diversity. Unlike Druck and Smith (2026), we also simulate this agentic setting.

Guided by a system prompt (Appendix C), the model decomposes the question, issues its own retrieve(query) calls against that question’s chunked retrieval store, and only then synthesizes an answer. It runs within the same recursive simulation as the other settings. Over 30 rounds, each round’s synthesized answer-document is embedded back into the store. Full implementation details are in Appendix D.

4 Experiment Setup

4.1 Datasets

We measure collapse in two settings: subjective entity comparison and multi-hop factual question answering.

Entity Comparison Dataset. First, we use a subset of the entity comparison dataset of Druck and Smith (2026), built from production prompt logs, curated keyword sets, and hand-authored open-domain queries. The corpus focuses primarily on non-factoid entity comparison queries of the form “*What are the best restaurants in the Bay Area?*”, which produce a list of entities with supporting details. This structure is particularly useful for studying answer diversity collapse by comparing the entity distribution of collapsed and non-collapsed responses. The subset we use contains 400 entity comparison questions.

HotpotQA Dataset. To measure whether collapse degrades task accuracy, we additionally consider HotpotQA (Yang et al., 2018), a multi-hop reasoning dataset requiring synthesis across documents. Unlike the entity comparison queries, its questions have short ground-truth answers, so accuracy can be scored directly. We use the first 1400 queries of the Dev set.

4.2 Evaluation Metrics

To quantify RAG collapse, we track response diversity and downstream task performance across simulation rounds. Let $A_r = (a_1, a_2, \dots, a_n)$ denote the list of $n = 10$ responses generated for a question in round r .

4.2.1 Diversity Metrics

Pairwise ROUGE-L: To capture lexical convergence, we compute the mean pairwise ROUGE-L F_1 over all pairs in A_r . Higher values indicate greater lexical overlap, and therefore less diversity, among responses.

Same-Answer Percentage: We also report the fraction of response pairs in a round judged to express the same core answer, evaluated by an LLM judge (Qwen2.5-7B-Instruct) over a fixed random sample of ten pairs per round (prompt in Appendix C). Higher values indicate greater semantic convergence among responses.

For entity comparison questions we perform additional analysis similar to prior work (Druck and Smith, 2026) that tracks how the distribution of mentioned entities evolves across simulation

rounds. Note that entity-level metrics are not reported for HotpotQA, whose answers are short factual spans rather than entity lists.

Unique Entities: Let $E(a_i)$ denote the set of canonical entities mentioned in response a_i . The number of distinct entities appearing across responses in round r is defined as

$$\text{UniqueEntities}_r = \left| \bigcup_{i=1}^n E(a_i) \right|$$

A decrease indicates that the range of entities being suggested by the model is narrowing.

Collapse Rates: For the Entity Comparison dataset, we call a simulation collapsed at round r if every response generated in that round mentions the same set of canonical entities (i.e., there is no entity variation across the round’s responses).

We then report, across questions, the percentage collapsed at the first round (**Collapsed at Start**), the percentage collapsed at the final round (**Collapsed at End**), and the mean fraction of rounds that are collapsed over the full simulation (**Rounds Collapsed**). Error bars on these quantities are 95% Wilson score intervals for a binomial proportion; start and end intervals are computed over the 400 questions, and the rounds-collapsed interval is pooled over all question–round cells.

4.2.2 Accuracy Metric (HotpotQA)

Token-Level F_1 : We report the standard SQuAD token-level F_1 (Rajpurkar et al., 2016) between each response and the ground-truth answer, under the usual normalization (lowercasing, punctuation and article removal), averaged over the round’s n responses.

4.3 Implementation Details

We evaluate four answer-generation LLMs: Qwen2.5-14B-Instruct, Mistral-7B-Instruct-v0.3, Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B, with the document generator fixed to Qwen2.5-7B-Instruct (Section 3). Both use temperature 0.7, so any convergence is attributable to the retrieval feedback loop rather than deterministic decoding. We generate $n = 10$ responses per question in each round. For Replace-One and Search, we randomly select one response to convert into the document fed into the next round. For Replace-All, we convert all 10. Retrieval settings are given in Section 3. Round budgets differ by contamination speed: 10 rounds for Replace-All, 20 for Replace-One, and 30 for Search.

Entity Extraction Pipeline: Entities are extracted from each response using GPT-5.2² prompted to return a JSON array of named entity mentions. Extracted mentions are then canonicalized with the same model, grouping variant references (e.g., “Sony XM5,” “WH-1000XM5”) under a single canonical name. This mapping is computed once per question over all rounds, producing a stable entity set against which per-round diversity is measured. Both prompts are in Appendix C. GPT-5.2 is used only as an evaluation instrument for entity extraction and canonicalization, not as part of the studied RAG system, which uses only open models.

5 RAG Collapse Results

5.1 Entity Dataset Results

We reproduce the collapse finding of Druck and Smith (2026) in an independent implementation with open models. As the context becomes increasingly AI-generated, fewer unique entities are mentioned, lexical variety decreases, and the resulting distribution no longer reflects the model’s round-1 distribution (Figure 2). For Qwen2.5-14B-Instruct, the fraction of questions collapsed by the final round is 66.8% (95% Wilson CI [62.0, 71.2]) under Replace-All and 59.8% [54.9, 64.4] under Replace-One, against 48.0% [43.1, 52.9] under Search, where self-generated and original documents compete during retrieval. Appendix A.1 (Figures 5, 6, and 7) shows the same pattern holds for the other models, most clearly under direct contamination (Replace-All and Replace-One).

Whether this narrowing is harmful depends on the question. For factual questions, concentrating on the correct answer is not harmful. Subjective questions have no single correct answer, and the narrowing we observe is not evidence-driven, as the same loop over unchanged references does not converge, leaving only sampling noise. Narrowing without new evidence concentrates exposure on fewer entities, which prior work identifies as a harm (Kleinberg and Raghavan, 2021).

5.2 HotpotQA Baseline Results

On HotpotQA, we evaluate F_1 across rounds for all four answer-generation models. Comparing F_1

²We found that for this task, frontier commercial models far outperformed open models. Open models extracted many false positive entities that inflated entity counts and suppressed collapse metrics.

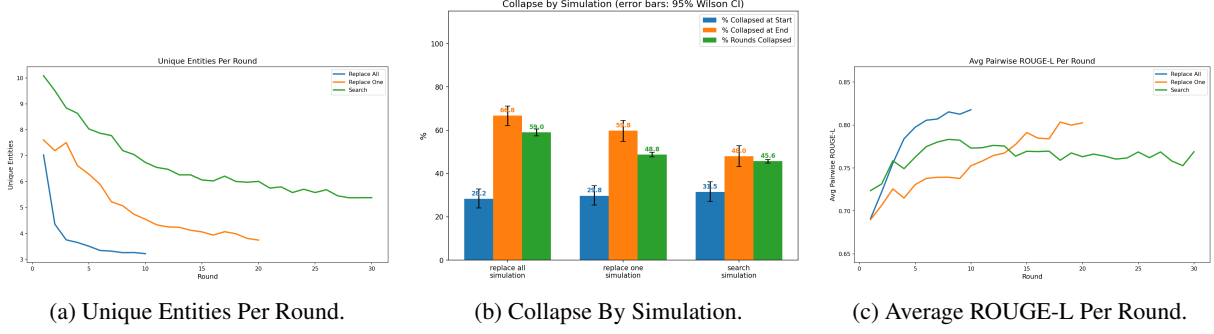


Figure 2: Entity Dataset results with Qwen2.5-14B-Instruct: as context becomes increasingly AI-generated, unique entities decline (a) and pairwise ROUGE-L rises (c), indicating loss of diversity and growing similarity between responses; panel (b) summarizes the fraction of collapsed simulations at the start and end of each run. Replace-All converges immediately, Replace-One follows a more gradual trajectory, and Search collapses most slowly as self-generated and original references compete.

in the first and final rounds (Appendix A.3, Table 1), only Qwen2.5-14B-Instruct declines significantly (ΔF_1 from -0.018 to -0.024 across simulations, 95% bootstrap CIs excluding zero). Mistral-7B-Instruct-v0.3 significantly *improves* ($+0.044$ to $+0.062$; Figure 29a), consistent with collapse onto a consensus answer that raises token overlap for a low-baseline model, while Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Qwen-7B show no consistent change (Appendix A.3). The dataset has 1400 questions, so these intervals are narrow, and the null results reflect small effects rather than insufficient statistical power.

For Qwen2.5-14B-Instruct, F_1 drops within the first few rounds under Replace-All and Replace-One, then plateaus (Figure 4a, which overlays the baseline and re-ranked curves). The decline under Search is more gradual. F_1 fluctuates across 30 rounds but trends downward, tracking the growing AI contamination of the retrieval pool, which rises from 0% to roughly 90% of retrieved references (Figure 4b).

HotpotQA answers are short factual spans, so response diversity is low initially and there is little room for collapse to cause an accuracy drop. Where F_1 does degrade, the document generator hallucinates entities when rewriting answers into articles and the answer generator conditions on these fabricated facts, so downstream harm is driven by drift rather than loss of diversity.

5.3 Agentic RAG Results

Agentic RAG consistently improves diversity metrics compared to all three baseline settings, while maintaining a similar AI reference percentage (Appendix A.1, Figure 8). For Qwen2.5-14B-Instruct,

the end-of-loop collapse rate is 27.2% (95% Wilson CI [23.1, 31.8]), against 66.8%, 59.8%, and 48.0% for Replace-All, Replace-One, and Search (Section 5). The comparison is not matched, however. Agentic RAG begins much lower, at 7.2% [5.1, 10.2] versus 28.2–31.5% for the baselines, and at approximately 1.01 tool calls per round (Figure 9) we cannot attribute the gains to iterative retrieval rather than to the richer system prompt. We treat this as promising but prompt-sensitive.

6 What Causes RAG Collapse?

The preceding sections establish that RAG collapse occurs. In this section, we analyze why.

Replace-All, the inference-time analog of the recursive training studied by Shumailov et al. (2024), collapses rapidly and severely (Section 5). The more informative settings are Replace-One and Search, where self-generated references compete with human references. Unless noted, we analyze Qwen2.5-14B-Instruct under Replace-One.

6.1 Collapse begins before full contamination

In Replace-One and Search, collapse is already underway while human references remain in the context. In Replace-One, one reference is replaced per round and the context becomes fully self-generated only at round 10, yet the number of unique words per round falls from 64.0 at round 1 to 51.3 at round 9 and pairwise ROUGE-L rises over the same interval, while roughly half of the context consists of original references through the middle rounds. Collapse is therefore not merely a consequence of the context becoming entirely self-generated. The self-generated references seem to have more influence.

6.2 Self-generated references are over-cited

This influence is visible in the citations. Following Druck and Smith (2026), we compute the ratio of a reference group’s citation share to its share of the context. At round 1, self-generated references are 12.0% of the context but receive 26.5% of citations under Replace-One (ratio 2.2; shares averaged over questions), and 14.8% versus 24.0% under Search (ratio 1.6): when the model’s own content first enters the pool, it is cited about twice as often as expected. Under Replace-One the context is constructed by slot replacement rather than retrieval, so this excess reflects selection among available references rather than a retrieval advantage. The attribution is a leave-one-out counterfactual rather than a raw overlap match: a reference is credited only when regenerating the answer with that reference removed changes the answer (full procedure in Appendix B). This over-citation is robust to the citation cutoff and to the attribution rule, is not explained away by a query-alignment control, and replicates across all four models and the agentic setting (Appendix B).

6.3 Over-citation is not fully explained by AI authorship

A natural explanation is that the model simply favors AI-generated text. We can test this because a substantial fraction of the original web references are themselves AI-generated: labeling the round-1 originals with GPTZero³, we find 30% are classified as AI. We separate references into self-generated, AI-written, and human-written. If AI authorship drove over-citation, the AI-written originals should be over-cited as strongly as self-generated ones. They are not: their mean citation rate is 0.123, far below the 0.281 of self-generated references, though above the 0.078 of human-written references (+0.045, $p < 0.001$). Over-citation tracks self-generation far more than AI authorship.

6.4 Collapse is not fully explained by style

AI-generated documents differ from human-written ones in both factual accuracy and stylistic regularity. AI text tends to have more uniform structure, heavier discourse marking, and lower sentence-level variability. Is collapse driven solely by the *stylistic uniformity* of AI text? To isolate style, we rewrite each round-1 original reference once, us-

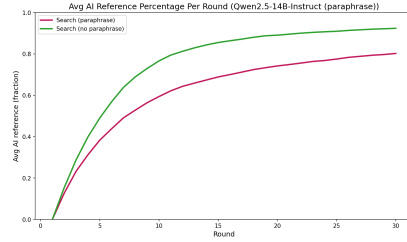


Figure 3: AI-reference percentage per round for Qwen2.5-14B-Instruct under paraphrasing across retrieval strategies. Under Search, the AI-reference share rises more slowly than in the unmitigated baseline, indicating that paraphrasing slows recursive self-reinforcement.

ing the same prompt we use to synthesize articles from responses (Appendix C), preserving its factual content while imposing AI-like style, then run the standard simulation unchanged. Full implementation details are in Appendix D.

Paraphrasing consistently reduces collapse severity for Qwen2.5-14B-Instruct (Figures 17 and 18, Appendix). The effect is clearest in the diversity metrics. Baseline entity diversity declines sharply across rounds but decays more slowly under paraphrasing. The effect is strongest under Search and Replace-One, which still interact with original references, and weakest under Replace-All, where self-generated references immediately replace the corpus.

Paraphrasing also slows recursive self-reinforcement: the AI-reference percentage grows more slowly, particularly under Search (Figure 3). This suggests that changing document phrasing alters retrieval trajectories enough to expose the generator to more varied evidence across rounds.

The same pattern holds across Mistral-7B-Instruct-v0.3, Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B (Appendix, Figures 20, 21, and 22), suggesting the effect generalizes across model families. If style alone caused collapse, equalizing it should stop it; instead collapse still occurs, only more slowly. Stylistic uniformity therefore accelerates collapse but cannot be its sole cause.

6.5 Over-citation is not fully explained by reference quality

Self-generated references may simply be better answers. To control for this, following Druck and Smith (2026), we score every round-1 reference along eight quality dimensions (relevance, accu-

³<https://gptzero.me>

racy, thoroughness, specificity, recency, organization, direct-answer, and overall) using GPT-5.2, and check whether the citation preference survives. Restricting to references scored maximally (5/5) on direct-answer, organization, and relevance, self-generated references are still cited 2.1 times as often as originals (0.295 versus 0.142). A linear model of citation rate on provenance, controlling for all eight dimensions, estimates a self-generation effect of +0.154 (95% bootstrap CI [+0.12, +0.19], $p < 0.001$), while the AI-written indicator is not significant (+0.004, $p = 0.65$). Quality does not fully account for over-citation.

6.6 Collapse is not fully explained by self-preference

Self-generated references are over-cited, and this is not fully explained by AI authorship, writing style, or reference quality. Is it a form of self-preference (Panickssery et al., 2024)? If so, references built from a different model’s answers should not cause collapse. We rerun all three simulations with Qwen2.5-14B-Instruct as the answer model, but build the synthetic references from a different model’s answers: Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, or DeepSeek-R1-Distill-Qwen-7B answers the same context, and Qwen2.5-7B-Instruct rewrites that answer into the document fed back into the loop. Collapse persists at a comparable rate for every source model, statistically indistinguishable from the same-model baseline. By the final round 59.0–71.2% of questions collapse across the three source models under Replace-All, against 66.8% for Qwen’s own answers, 50.8–65.5% against 59.8% under Replace-One, and 45.2–47.8% against 48.0% under Search (Appendix A.2). Self-preference is therefore not the primary driver: the loop collapses even when the documents derive from another model’s answers. Documents synthesized from any model’s answer may share features of LLM synthesis that the answer model favors; whether this signature is family-specific is ongoing work.

7 Mitigation Methods

7.1 Re-Ranker Mitigation

This strategy extends Search with a re-ranking step between retrieval and generation that penalizes likely-AI documents, so human-authored content is preferred when relevance is comparable.

Given a query, we retrieve the top- n documents

($n = 20$) by dense retrieval, min-max normalize their scores to $[0, 1]$, and estimate each document’s AI-generation probability $P_{\text{AI}}(D)$ with an AI-content detector. We use the soft probability rather than a hard filter, since noisy detectors would otherwise discard misclassified human documents. Each retrieval score is adjusted as:

$$s' = s \cdot (1 - \lambda \cdot P_{\text{AI}}(D)) \quad (1)$$

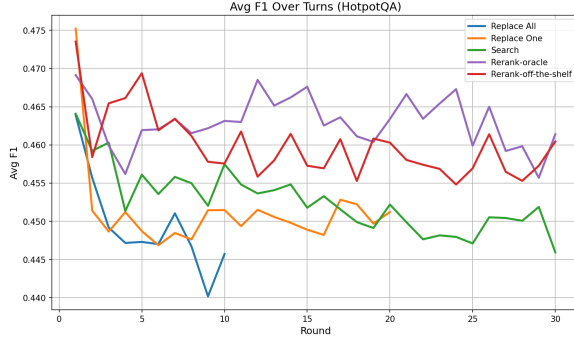
where s is the original normalized retrieval score and $\lambda \in [0, 1]$ is a penalization strength hyperparameter. The documents are then re-sorted by s' and the top- k results ($k = 10$) are passed to the generator (Figure 23, Appendix).

For the AI detection model we use desklib/ai-text-detector-v1.01, built on the DeBERTa (He et al., 2021) encoder architecture. The re-ranker scores every retrieved document at every round, so it needs a local model rather than the commercial detector used in Section 6.3. Because a substantial fraction of original references in the Entity Dataset are themselves AI-generated and collapse is driven by self-generated content rather than AI authorship alone, an off-the-shelf AI-content detector may be overly aggressive. We therefore also evaluate an oracle detector, an upper bound on a detector that perfectly identifies the contaminating content. It assigns P_{AI} sampled uniformly from $[0.95, 1.0]$ to self-generated documents and from $[0.0, 0.05]$ to originals, giving near-perfect class separation with minor stochastic variation in ranking.

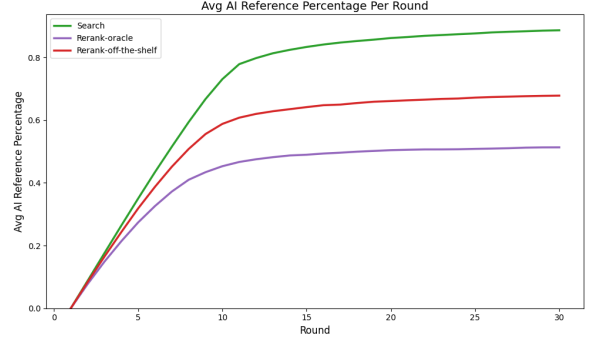
For our experiments, we use the desklib detector without any fine-tuning. We set $\lambda = 0.7$ for the main comparison and additionally sweep $\lambda \in \{0.1, 0.5, 0.7\}$ for Qwen2.5-14B-Instruct (Appendix A.1, Figure 24).

7.2 GEPA Prompt Optimization

The RAG system prompt is hand-crafted and held fixed across conditions. Optimizing it directly against collapse dynamics could improve robustness without architectural changes. We apply GEPA (Agrawal et al., 2025) (Genetic-Pareto), which uses an LLM (Claude Opus 4.1) to reflect on execution traces and mutate prompt candidates toward a Pareto frontier. Seeded from the baseline, each candidate is scored on an entity-based anti-collapse score and a quality score judged by GPT-4o, across all three simulations (50 HotpotQA and 50 Entity Comparison questions); details in Appendix D.



(a) F_1 across iterations for baselines and rerank mitigation.



(b) AI reference % per round (Search + rerank mitigation).

Figure 4: Re-ranker mitigation on HotpotQA with $\lambda = 0.7$, Qwen2.5-14B-Instruct. (a) off-the-shelf detector and oracle detector settings compared against baselines. (b) the proportion of AI-generated documents in the retrieved context over rounds.

8 Mitigation Results

We find that collapse is robust to our mitigations.

Re-Ranker Results *HotpotQA*. On HotpotQA, the oracle re-ranker yields at most modest improvements for Qwen2.5-14B-Instruct: F_1 remains slightly closer to the initial non-collapsed level than the unmitigated Search baseline across 30 rounds, though the gap is small (Figure 4a).

Entity Comparison. On the Entity Comparison dataset, re-ranking fails to prevent collapse, and the oracle detector performs at most marginally better than the off-the-shelf detector. With either detector at $\lambda = 0.7$, entity diversity declines across rounds at a rate close to the unmitigated Search baseline, and pairwise ROUGE-L rises similarly, indicating comparable convergence among responses (Appendix A.1, Figure 25). The same oracle-vs-desklib comparison for Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, and DeepSeek-R1-Distill-Qwen-7B (Figures 26, 27, and 28) shows the same pattern.

Re-ranking can only reorder the candidates that retrieval returns. As self-generated documents accumulate, they outrank original references on similarity to the query, so the retrieved pool becomes increasingly self-generated. By the final round, even the oracle detector, which down-ranks every self-generated document, retrieves on average only one original document out of ten. At that point there is nothing left to re-rank, and no detector can help. As described in Section 5, running the simulations using original references does not cause collapse. Therefore, mitigating collapse requires filtering out self-generated references before retrieval.

GEPA Prompt Optimization Results GEPA converged immediately, with the baseline prompt already on or near the Pareto frontier; across all three variants, no candidate improved either score over the baseline. We read this as inconclusive rather than negative: at ten rounds per variant the signal is likely too noisy, and the objective may not penalize the answer-derived preference that drives collapse. Longer horizons and stricter anti-collapse objectives are natural next steps.

9 Conclusion

As AI-generated documents accumulate in the retrieval corpus, outputs become less diverse and, for some models, less accurate. Our analysis traces this to an answer-derived preference, in which the model conditions disproportionately on documents derived from model answers. AI authorship, writing style, answer quality, and self-preference do not account for it. None of the mitigations we tried reliably prevent collapse. Paraphrasing and agentic retrieval slow it, re-ranking is not effective, and GEPA prompt optimization did not improve over the baseline. Because re-ranking can only reorder what retrieval returns, preventing collapse will likely require filtering self-generated references before retrieval and directly countering the answer-derived preference.

Distractor-based document synthesis, retrieval-stage filtering, AI-generated documents written for other queries, and diversity-preserving prompts are the focus of ongoing work.

Limitations

Our strongest evidence concerns diversity collapse. The downstream task-performance findings

are more limited in scope, and F_1 on HotpotQA moves in a model-dependent direction. It declines for Qwen2.5-14B-Instruct, rises for Mistral-7B-Instruct-v0.3, and is flat for Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Qwen-7B (Appendix A.3). We therefore cannot claim general downstream harm across model families. Where diversity collapse coincides with convergence onto a correct or common answer, token-overlap F_1 can even rise. A document-synthesis design that deliberately injects plausible misinformation (distractor-style), rather than condensing answers, is a more direct test of downstream harm, and one we are currently pursuing.

In our simulations, each question is handled separately. A question’s retrieval store starts with that question’s original references, and the only documents added to it come from answers to that same question. This isolates the retrieval feedback loop, but it does not model the open web, where most AI-generated documents were written for other queries. Our conclusions are therefore about this controlled setting. Testing a shared corpus of AI documents not written for the evaluation query is ongoing work. Round budgets also differ across simulations (10, 20, and 30 rounds for Replace-All, Replace-One, and Search) because contamination accumulates at different speeds, so cross-simulation comparisons of final collapse rates compare different amounts of elapsed contamination.

We find over-citation for all four models under Replace-One, even when we control for query alignment (Appendix B.3). In the Search simulation the effect does not hold: it is significantly positive for only one model, null for two, and significantly negative for Qwen2.5-14B.

We do not validate extraction against human annotation. Inaccurate extractions primarily produce false-positive entities that inflate counts and suppress collapse metrics, so poor extraction would make our results conservative. Pairwise ROUGE-L requires no extraction and shows the same trends.

Ethics Statement

This work involves no human subjects, no crowdsourcing, and no collection of personal data. The Entity Comparison dataset we use is derived from production prompt logs, curated keyword sets, and manually authored open-domain queries. It contains no personally identifiable information. Our dataset and our simulation and evaluation code are

publicly available. HotpotQA is a publicly available benchmark used under its existing terms.

Our re-ranker mitigation uses a third-party, off-the-shelf AI-text detector ([desklib/ai-text-detector-v1.01](https://github.com/desklib/ai-text-detector-v1.01)), which we did not develop. We also use GPTZero to label the original references in Section 6.3. Like all such detectors, they are imperfect and unreliable for high-stakes determinations of authorship, and we use them only as a research component, not as a deployable provenance tool.

We used AI assistance in writing portions of the manuscript and the experimental code, and AI models also serve as components of our method and evaluation (synthetic document generation, entity extraction and canonicalization, and LLM-as-a-judge scoring). All research questions, experimental design, analysis, and conclusions are the authors’ own. Our study examines a failure mode, RAG collapse under accumulating AI-generated web content, that bears directly on the trustworthiness of deployed retrieval systems. We believe characterizing it supports safer deployment. This work is consistent with the ACL Code of Ethics.

Acknowledgments

We thank the anonymous GroundLM reviewers for their constructive feedback. The computational resources for this work were provided by the Unity Research Computing Platform, a multi-institutional cluster led by the University of Massachusetts and the University of Rhode Island. This research received no external funding, and the authors declare no conflicts of interest.

References

- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2025. [GEPA: Reflective prompt evolution can outperform reinforcement learning](#). *Preprint*, arXiv:2507.19457.
- Gregory Druck and Ethan Smith. 2026. [RAG collapse: LLM responses collapse when retrieved documents are self-authored](#). *Preprint*, arXiv:2608.22118.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. [DeBERTa: Decoding-enhanced BERT with disentangled attention](#). *Preprint*, arXiv:2006.03654.

- Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2017. [Billion-scale similarity search with GPUs](#). *Preprint*, arXiv:1702.08734.
- Jon Kleinberg and Manish Raghavan. 2021. [Algorithmic monoculture and social welfare](#). *Proceedings of the National Academy of Sciences*, 118(22):e2018340118.
- Muhammad Ahmed Mohsin, Muhammad Umer, Ahsan Bilal, Zeeshan Memon, Muhammad Ibtsaam Qadir, Sagnik Bhattacharya, Hassan Rizwan, Abhiram R. Gorle, Maahe Zehra Kazmi, Nukhba Amir, Ali Subhan, Muhammad Usman Rafique, Zihao He, Pulkit Mehta, Muhammad Ali Jamshed, and John M. Cioffi. 2025. [On the fundamental limits of LLMs at scale](#). *Preprint*, arXiv:2511.12869.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jose Luis Paredes, Gregory Druck, Bevin Benson, and Ethan Smith. 2026. AI now writes as many online articles as humans. <https://graphite.io/five-percent/ai-now-writes-as-many-online-articles-as-humans-do>. Graphite. Accessed: 2026-06-08.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. [SQuAD: 100,000+ questions for machine comprehension of text](#). In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, Austin, Texas.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-BERT: Sentence embeddings using siamese BERT-networks](#). *Preprint*, arXiv:1908.10084.
- Ilya Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. [AI models collapse when trained on recursively generated data](#). *Nature*, 631(8022):755–759.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2024. [Text embeddings by weakly-supervised contrastive pre-training](#). *Preprint*, arXiv:2212.03533.
- Tianyu Wang, Akira Horiguchi, Lingyou Pang, and Carey E. Priebe. 2025. [LLM web dynamics: Tracing model collapse in a network of LLMs](#). *Preprint*, arXiv:2506.15690.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. [HotpotQA: A dataset for diverse, explainable multi-hop question answering](#). *Preprint*, arXiv:1809.09600.

A Additional Results

A.1 Additional RAG Collapse Results: Entity Dataset

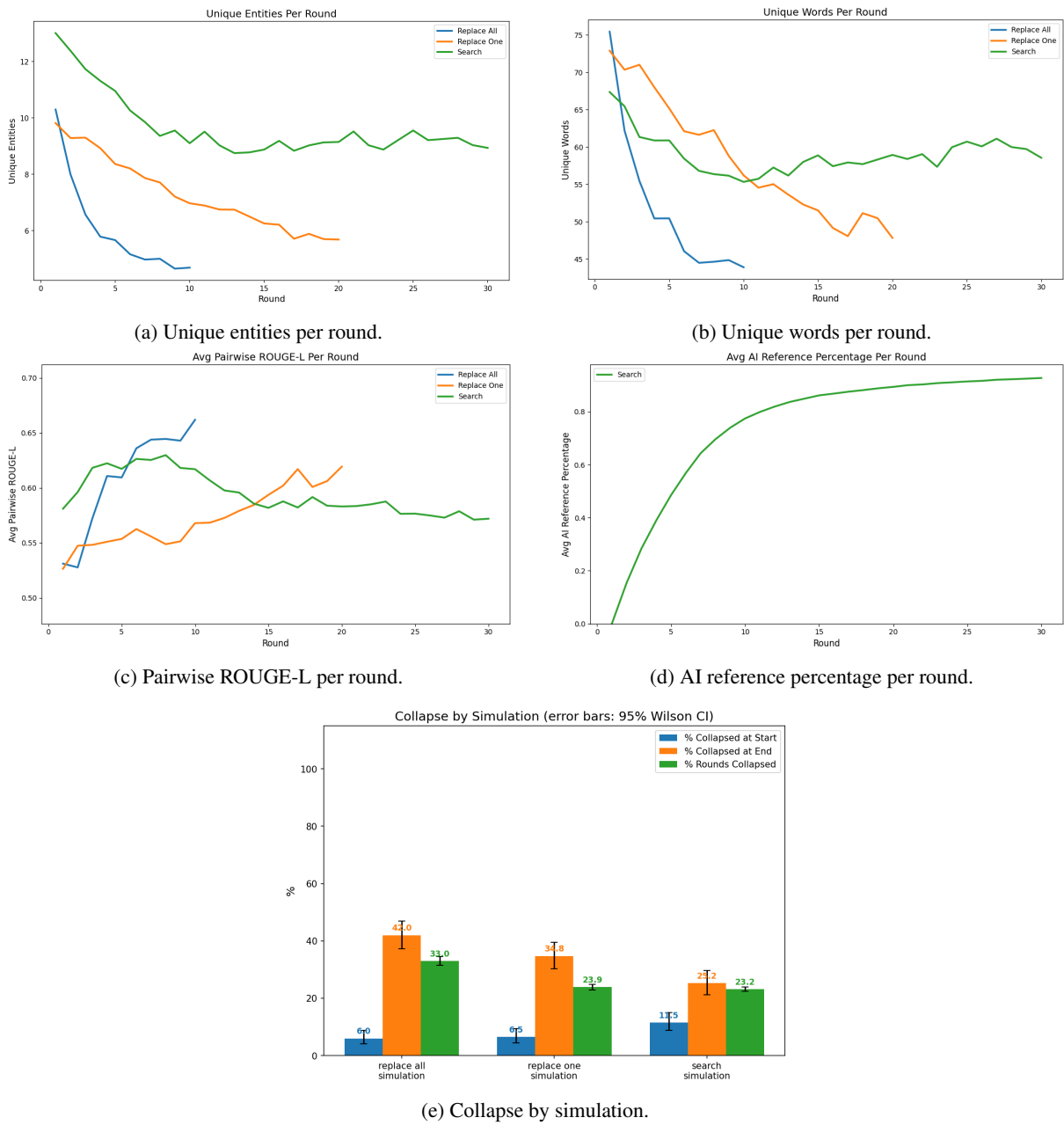


Figure 5: Entity Dataset results with Llama-3.1-8B-Instruct.

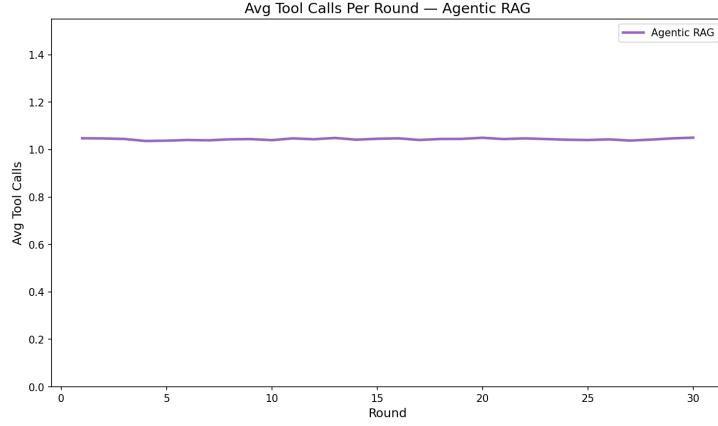


Figure 9: Average number of retrieve tool calls issued per round by the Agentic RAG model. The near-constant value of approximately 1.01 across rounds indicates that the model rarely issued more than one retrieval call per query, suggesting that diversity gains may stem from retrieval instruction design rather than iterative multi-step retrieval.

A.2 Cross-Model Collapse: Self-Preference Control

Section 6 reports that collapse persists when the documents fed back into the loop are built from a different model’s answers. Here we give the full per-source figures, using the entity-collapse metric of Section 4.2 over the 400-question Entity Comparison dataset, tagged identically to the baseline runs. Figure 10 overlays the same-model baseline (documents built from Qwen’s own answers) against the three source models; the baseline sits within the cross-model spread in every simulation. Figure 11 shows the per-round trajectories, and Figures 12–14 give the per-source detail.

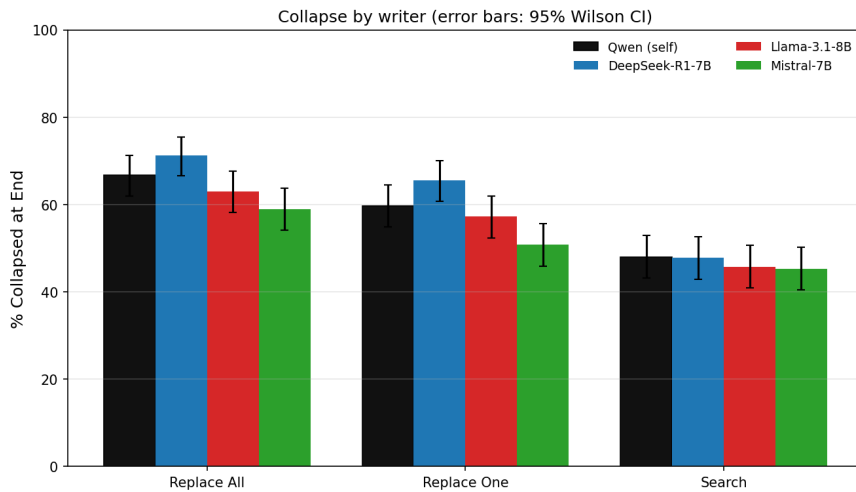


Figure 10: End-round collapse rate when the documents fed back into the loop are built from Qwen2.5-14B-Instruct’s own answers (outlined) versus from each source model’s answers, per simulation. Qwen2.5-14B-Instruct is the answer model throughout, and Qwen2.5-7B-Instruct rewrites each answer into a document. The same-model baseline is statistically indistinguishable from the source models (overlapping 95% Wilson CIs, 400 questions).

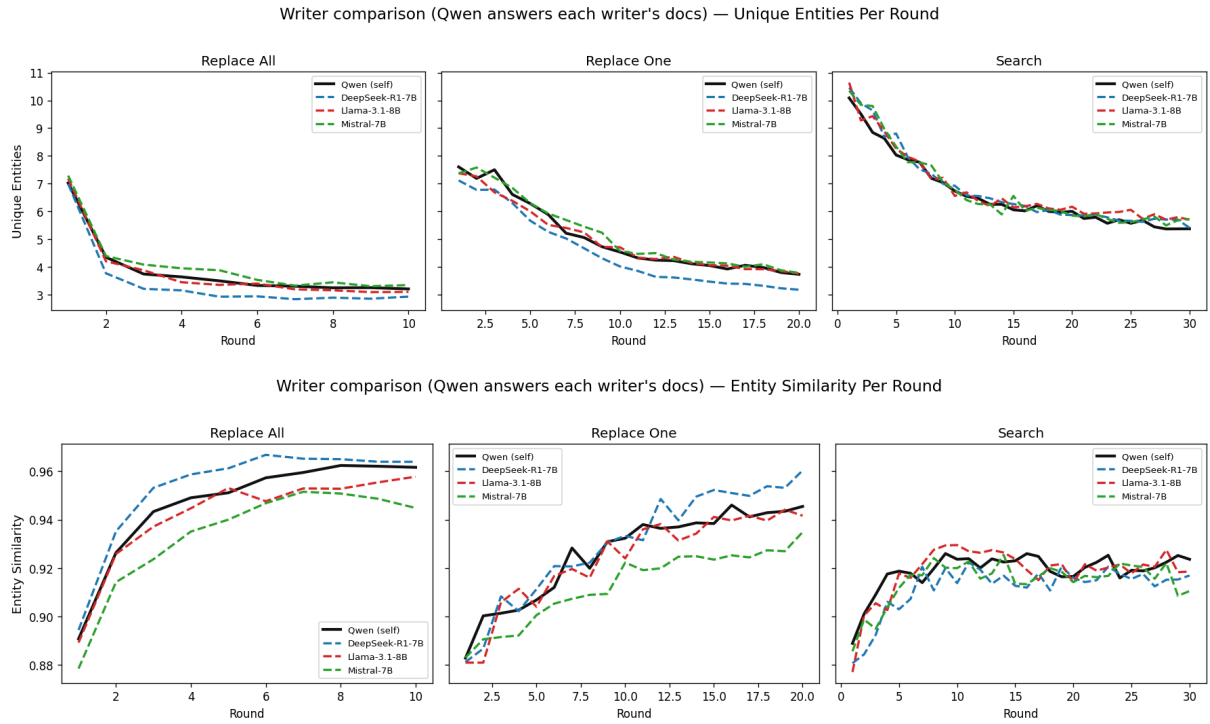


Figure 11: Per-round unique entities (top) and entity similarity (bottom) for Qwen2.5-14B-Instruct when the fed-back documents are built from each source model’s answers, per simulation (Replace-All / Replace-One / Search). Trajectories are near-identical across source models, including Qwen’s own answers.

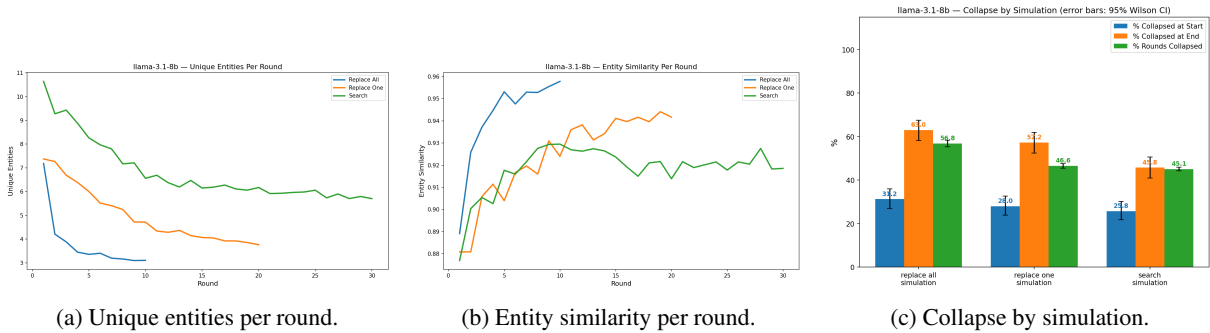


Figure 12: Cross-model entity metrics, Qwen2.5-14B-Instruct answering documents built from Llama-3.1-8B-Instruct’s answers, across Replace-All / Replace-One / Search.

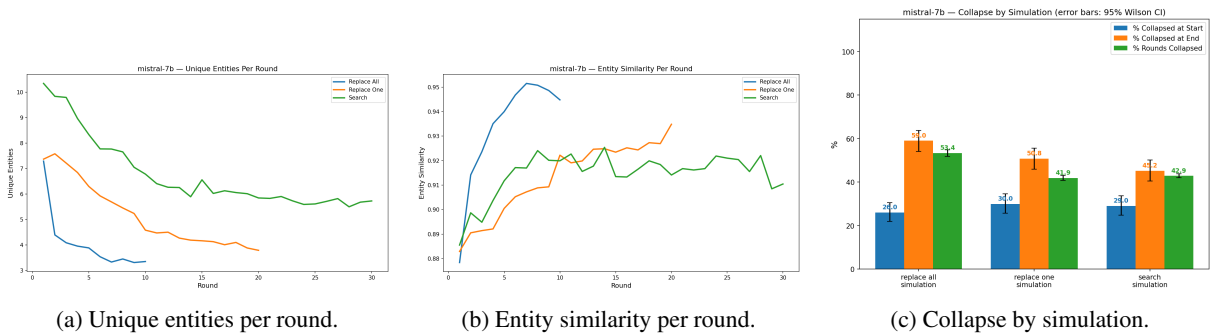


Figure 13: Cross-model entity metrics, Qwen2.5-14B-Instruct answering documents built from Mistral-7B-Instruct-v0.3’s answers, across Replace-All / Replace-One / Search.

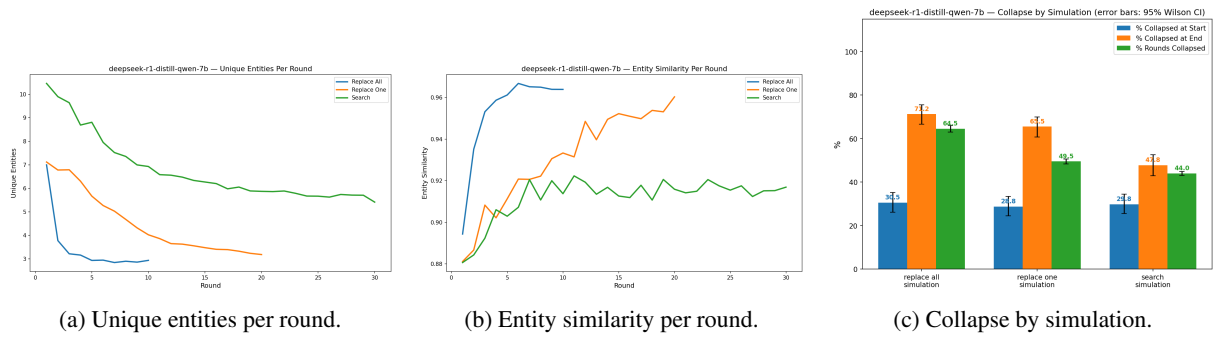
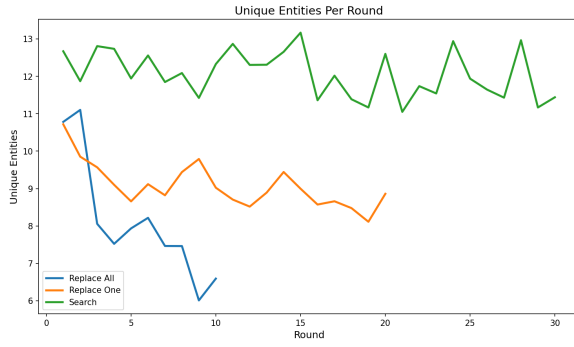
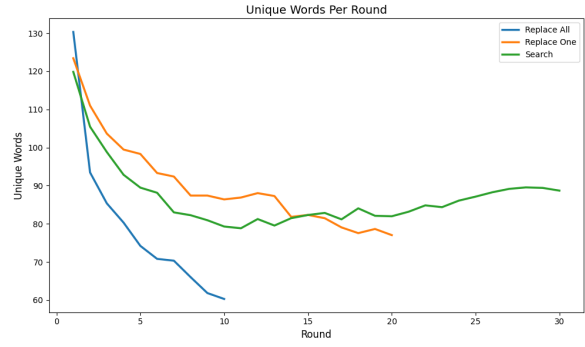


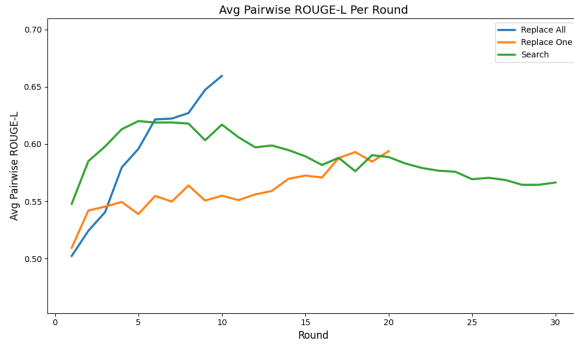
Figure 14: Cross-model entity metrics, Qwen2.5-14B-Instruct answering documents built from DeepSeek-R1-Distill-Qwen-7B's answers, across Replace-All / Replace-One / Search.



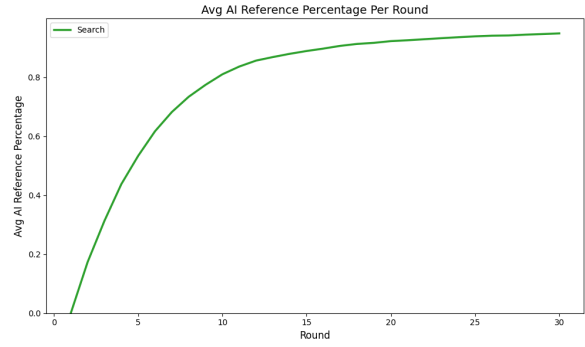
(a) Unique entities per round.



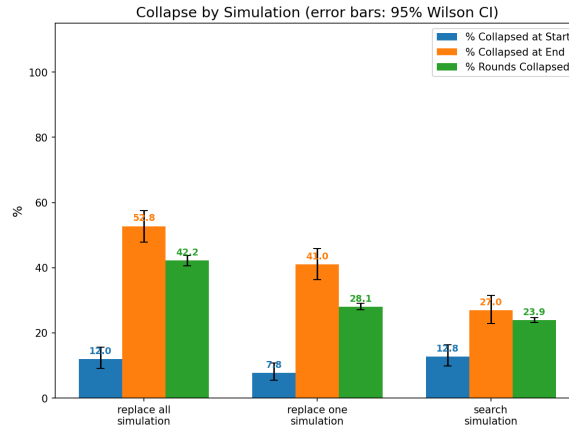
(b) Unique words per round.



(c) Pairwise ROUGE-L per round.

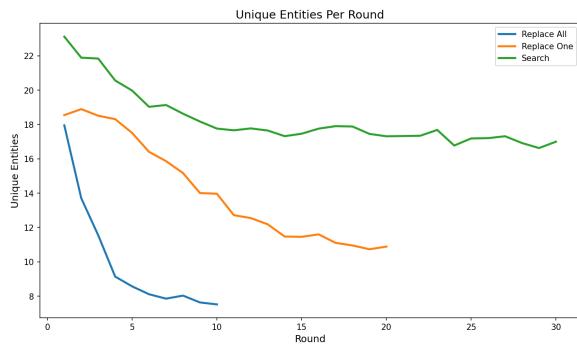


(d) AI reference percentage per round.

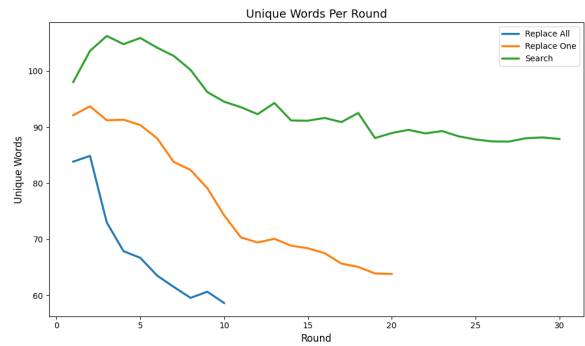


(e) Collapse by simulation.

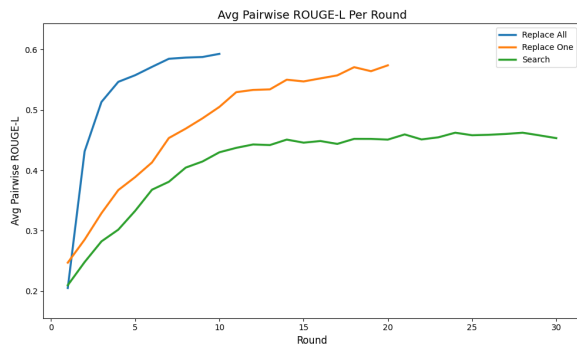
Figure 6: Entity Dataset results with Mistral-7B-Instruct-v0.3.



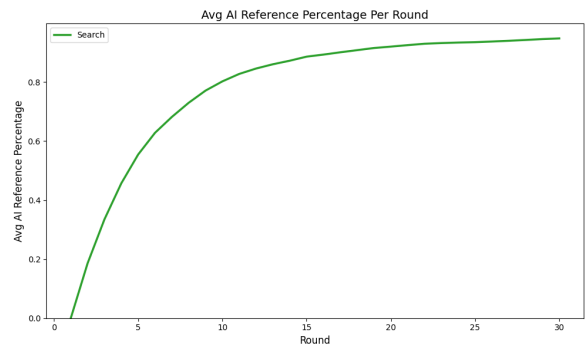
(a) Unique entities per round.



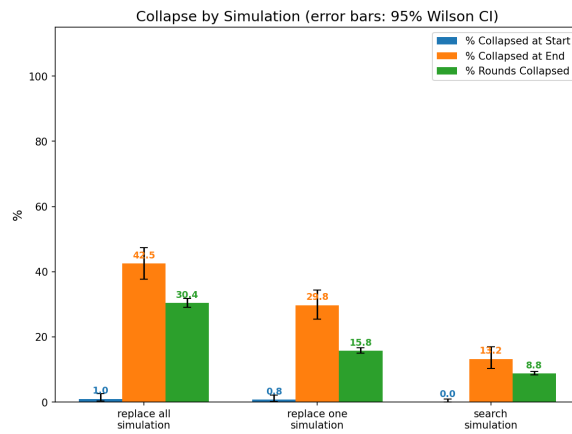
(b) Unique words per round.



(c) Pairwise ROUGE-L per round.

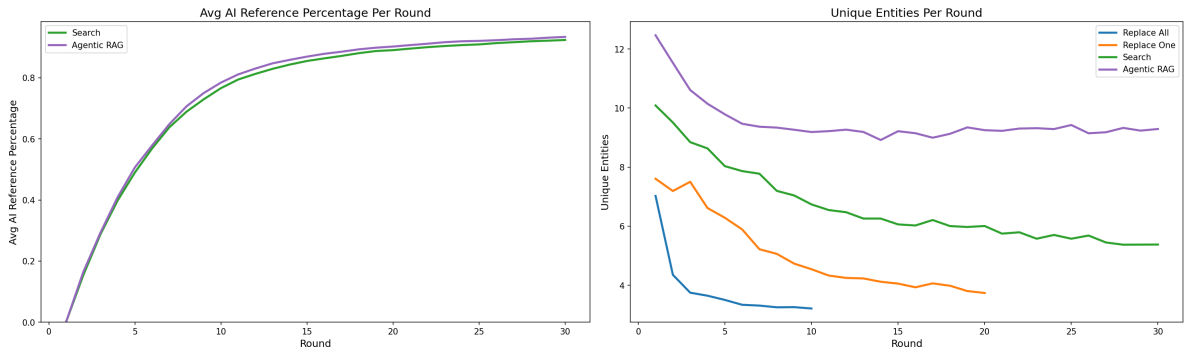


(d) AI reference percentage per round.



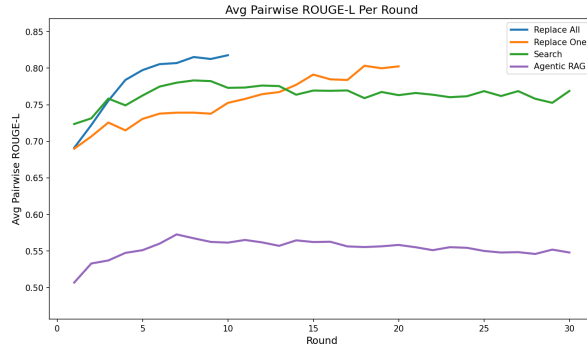
(e) Collapse by simulation.

Figure 7: Entity Dataset results with DeepSeek-R1-Distill-Qwen-7B.



(a) Average AI Reference Percentage Per Round.

(b) Unique Entities Per Round.



(c) Average Pairwise ROUGE-L Per Round.

Figure 8: Agentic RAG diversity and similarity metrics on the Entity Dataset with Qwen2.5-14B-Instruct. Agentic RAG preserves higher unique entity counts and lower pairwise ROUGE-L across rounds compared to all three baseline simulation settings, despite accumulating a similar proportion of AI-generated references over time.

A.3 Additional RAG Collapse Results: HotpotQA

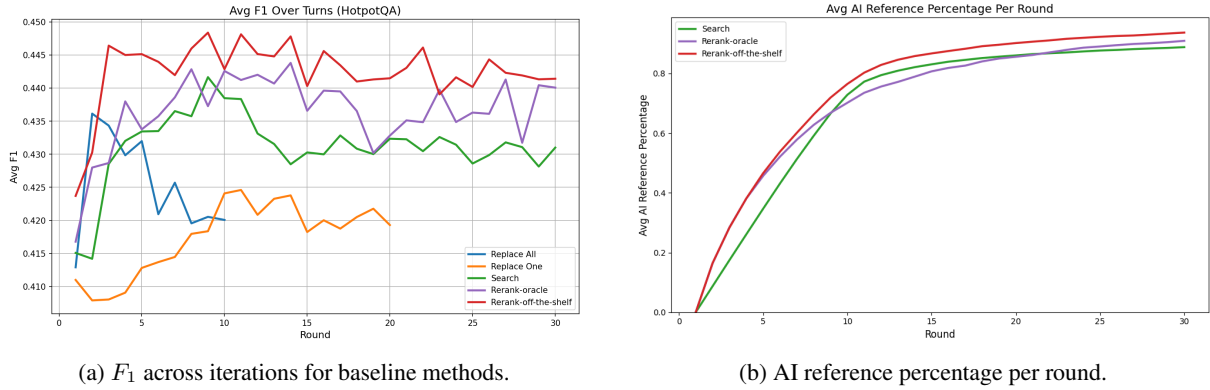


Figure 15: HotpotQA baseline results for Llama-3.1-8B-Instruct. (a) Average F_1 over rounds across simulation settings. (b) Proportion of AI-generated documents in the retrieved context.

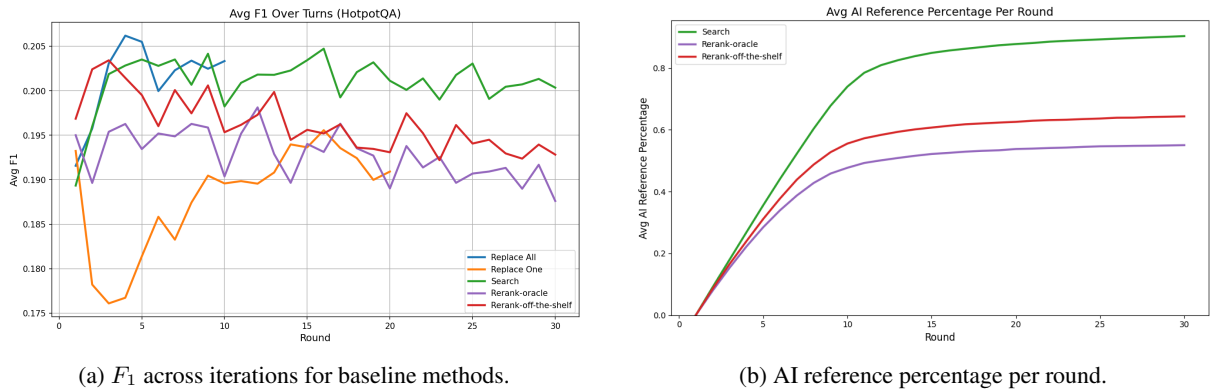


Figure 16: HotpotQA baseline results for DeepSeek-R1-Distill-Qwen-7B. (a) Average F_1 over rounds across simulation settings. (b) Proportion of AI-generated documents in the retrieved context.

With the current methodology, collapse does not consistently affect downstream task performance across all model families. Extending the simulation to use distractor-style document synthesis (where the synthesizer introduces plausible misinformation) would more directly test how retrieval contamination affects factual correctness.

Per-model effect sizes. Table 1 reports the round-0-to-final change in token-level F_1 per model and simulation, with 95% bootstrap confidence intervals resampled over the 1400 questions.

Model	Replace-All		Replace-One		Search	
Qwen2.5-14B	-0.018	[-0.034, -0.002]	-0.024	[-0.041, -0.007]	-0.018	[-0.034, -0.002]
Mistral-7B	+0.062	[+0.050, +0.076]	+0.044	[+0.031, +0.058]	+0.058	[+0.044, +0.072]
Llama-3.1-8B	+0.007	[-0.009, +0.024]	+0.008	[-0.009, +0.025]	+0.016	[-0.001, +0.033]
DeepSeek-R1-7B	+0.012	[+0.001, +0.023]	-0.002	[-0.013, +0.008]	+0.011	[-0.000, +0.022]

Table 1: Round-0 to final-round change in token-level F_1 on HotpotQA (mean over 1400 questions, 95% bootstrap CIs over questions). Only Qwen2.5-14B-Instruct declines significantly; Mistral-7B-Instruct-v0.3 significantly improves; Llama and DeepSeek show intervals spanning zero. The narrow intervals indicate the nulls are small genuine effects rather than low power.

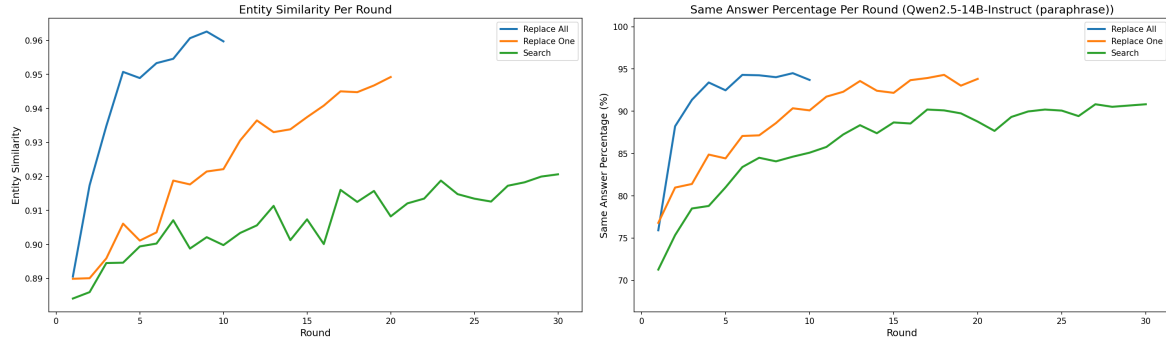


Figure 19: Additional paraphrasing metrics for Qwen2.5-14B-Instruct across retrieval strategies.

A.4 Additional Paraphrasing Results

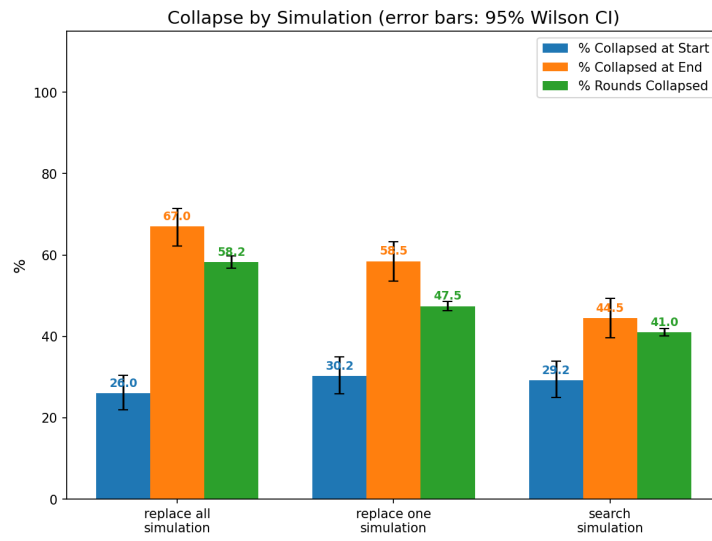


Figure 17: Collapse statistics for Qwen2.5-14B-Instruct under paraphrasing across retrieval strategies.

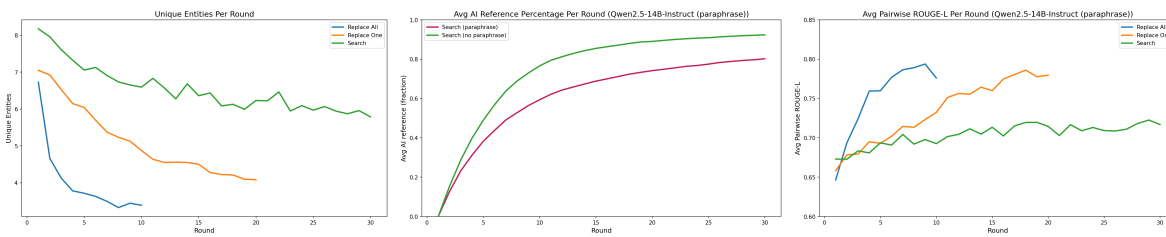


Figure 18: Paraphrasing results for Qwen2.5-14B-Instruct across retrieval strategies. Shown are unique entities per round, average AI-reference percentage per round, and average pairwise ROUGE-L per round. Compared to the baseline condition, paraphrasing slows lexical convergence, preserves entity diversity, and reduces recursive self-reinforcement.

Across Qwen2.5-14B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, and DeepSeek-R1-Distill-Qwen-7B (Figures 19, 20, 21, and 22), paraphrasing slows lexical convergence, preserves entity diversity, and reduces recursive self-reinforcement across retrieval strategies. It delays convergence but does not eliminate long-term homogenization once recursively generated content dominates the retrieval corpus.

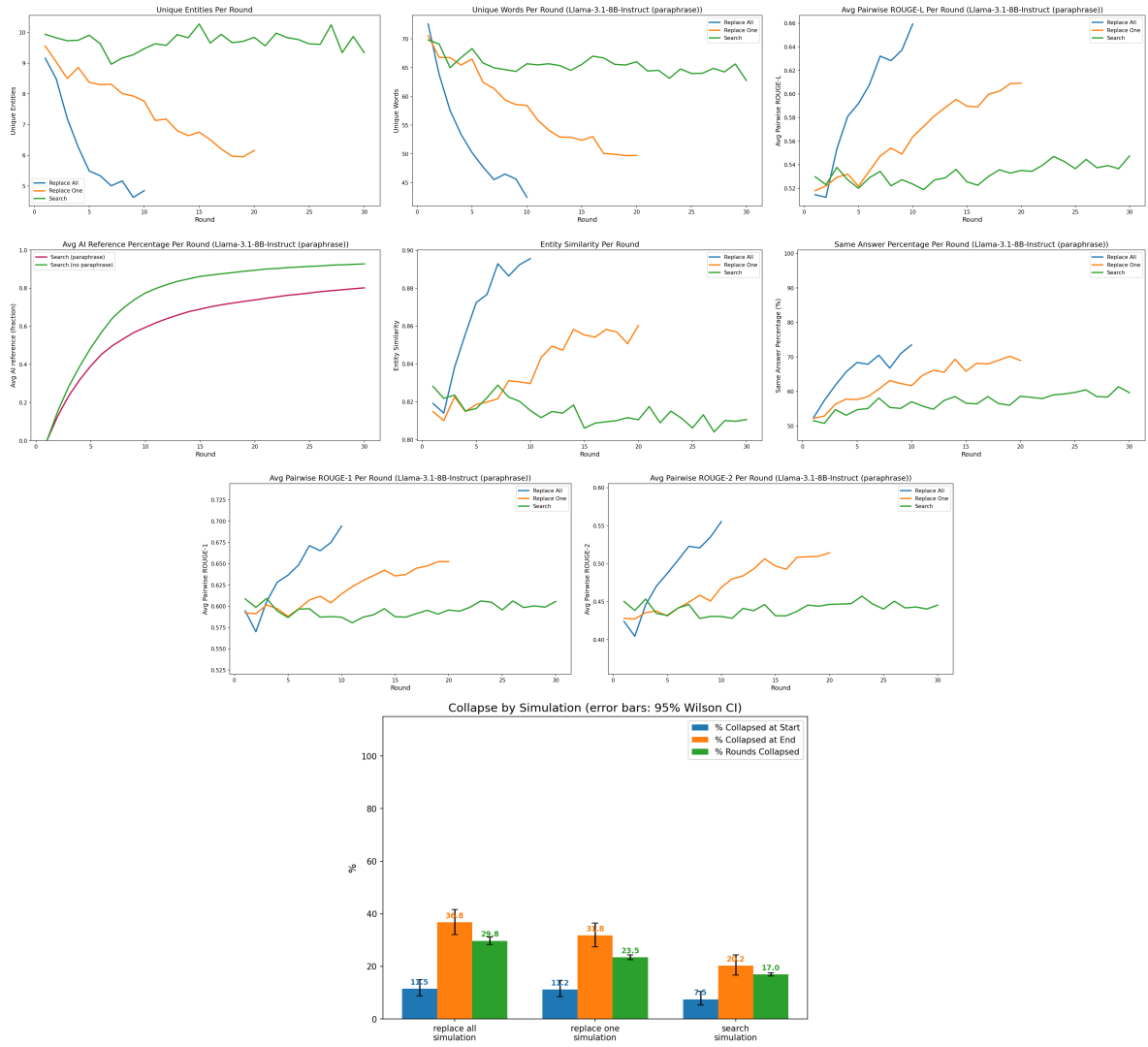


Figure 20: additional paraphrasing metrics for Llama-3.1-8B-Instruct across retrieval strategies: unique entities, unique words, pairwise ROUGE-L, AI-reference percentage, entity similarity, same-answer percentage, pairwise ROUGE-1 and ROUGE-2, and collapse by simulation.

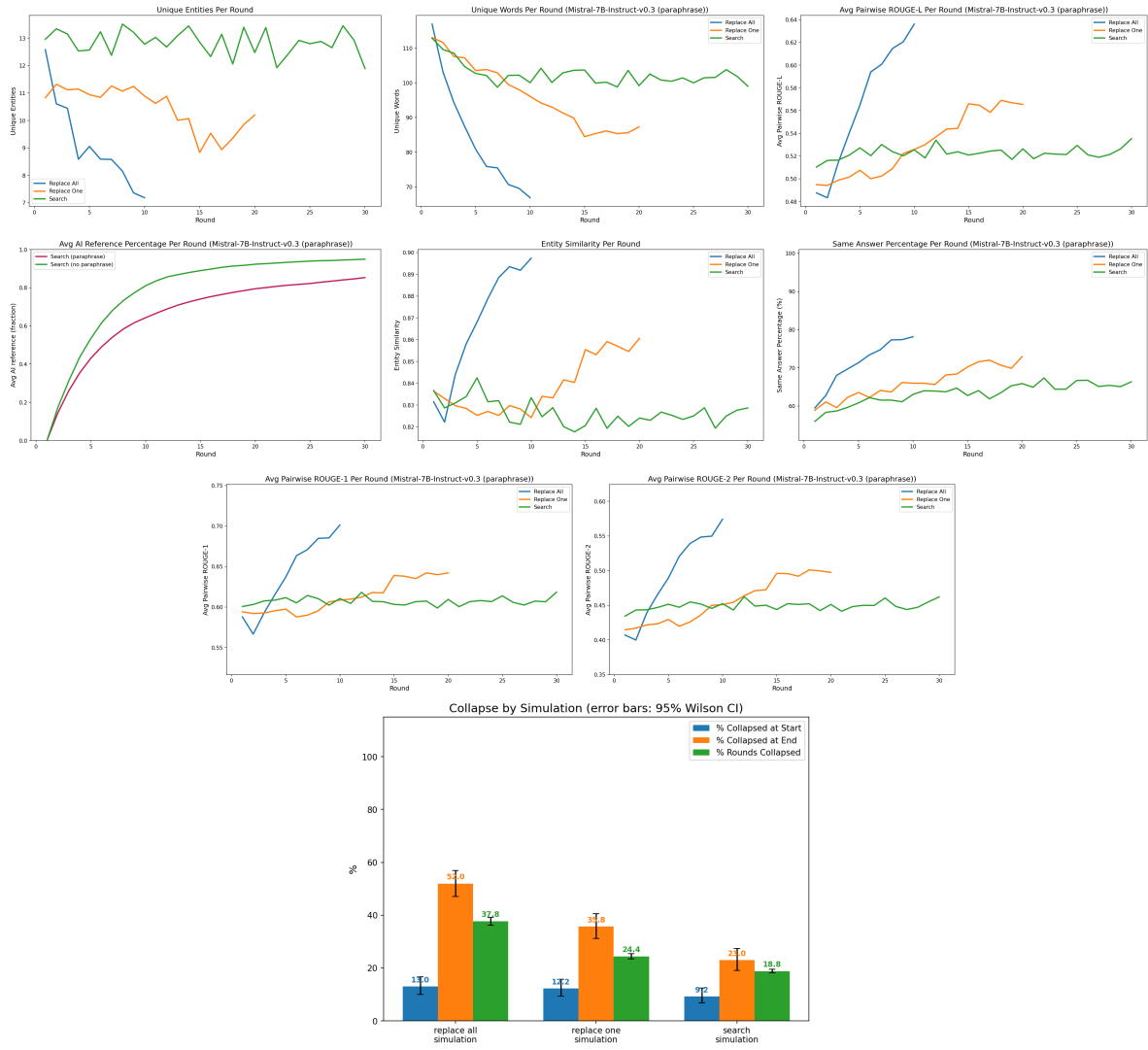


Figure 21: Additional paraphrasing metrics for Mistral-7B-Instruct-v0.3 across retrieval strategies: unique entities, unique words, pairwise ROUGE-L, AI-reference percentage, entity similarity, same-answer percentage, pairwise ROUGE-1 and ROUGE-2, and collapse by simulation.

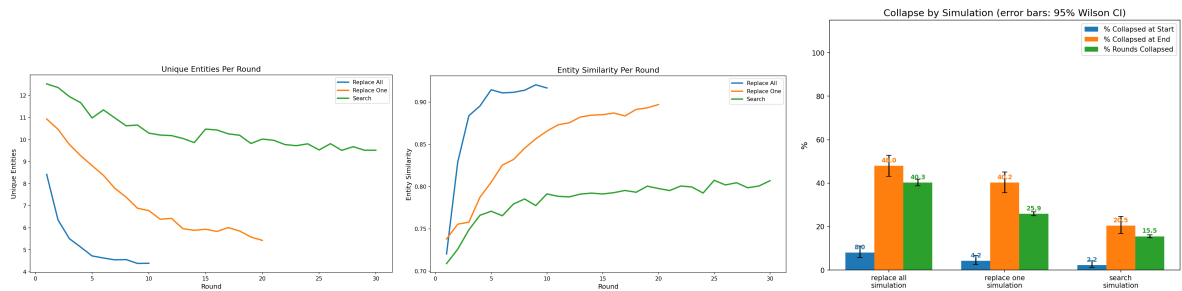


Figure 22: Additional paraphrasing metrics for DeepSeek-R1-Distill-Qwen-7B across retrieval strategies.

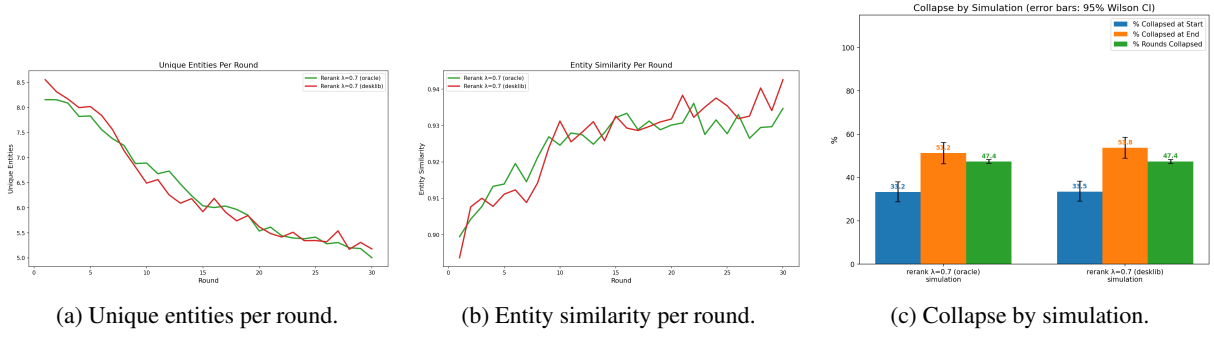


Figure 25: Re-ranker mitigation on the Entity Comparison dataset, Qwen2.5-14B-Instruct: oracle vs. off-the-shelf (desklib) detector at $\lambda=0.7$.

A.5 Additional Mitigation Results

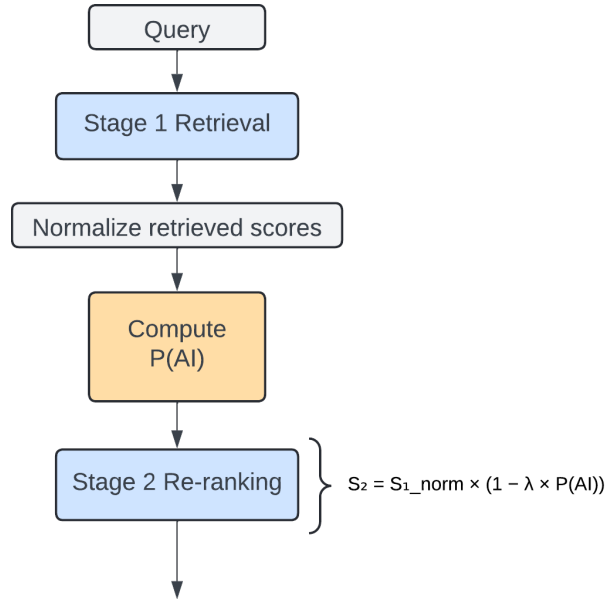


Figure 23: Overview of the Rerank pipeline. In the diagram, S_1 denotes the normalized retrieval score s and S_2 the adjusted score s' of Eq. (1); the re-ranking step applies $s' = s \cdot (1 - \lambda \cdot P_{AI}(D))$.

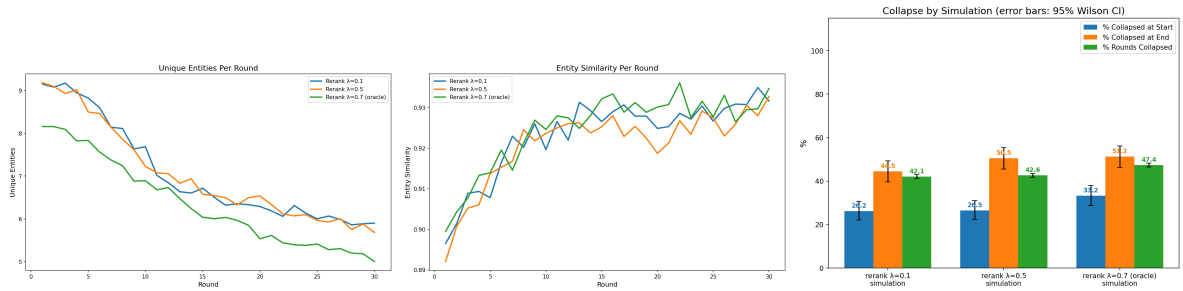


Figure 24: Re-ranker λ sweep ($\lambda = 0.1/0.5/0.7$, oracle detector) on the Entity Comparison dataset, Qwen2.5-14B-Instruct.

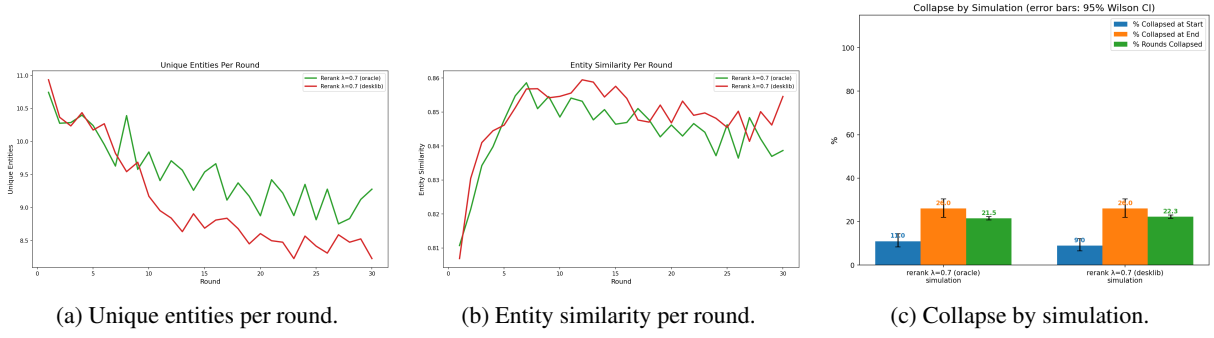


Figure 26: Re-ranker mitigation on the Entity Comparison dataset, Llama-3.1-8B-Instruct: oracle vs. off-the-shelf (desklib) detector at $\lambda=0.7$.

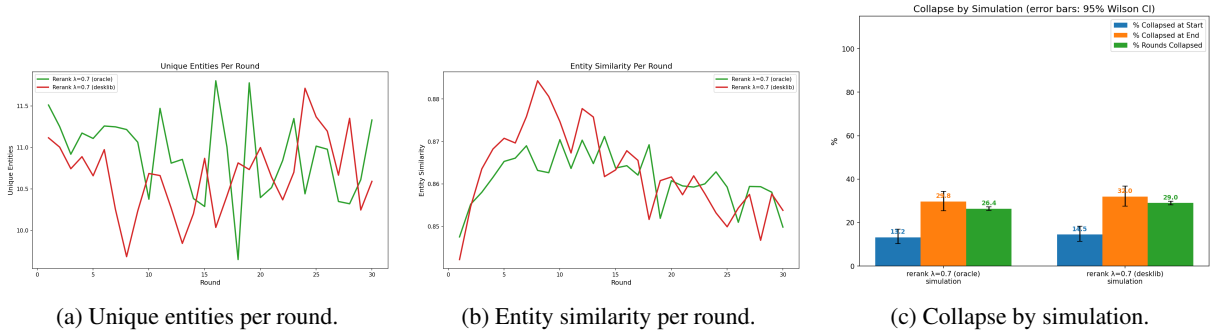


Figure 27: Re-ranker mitigation on the Entity Comparison dataset, Mistral-7B-Instruct-v0.3: oracle vs. off-the-shelf (desklib) detector at $\lambda=0.7$.

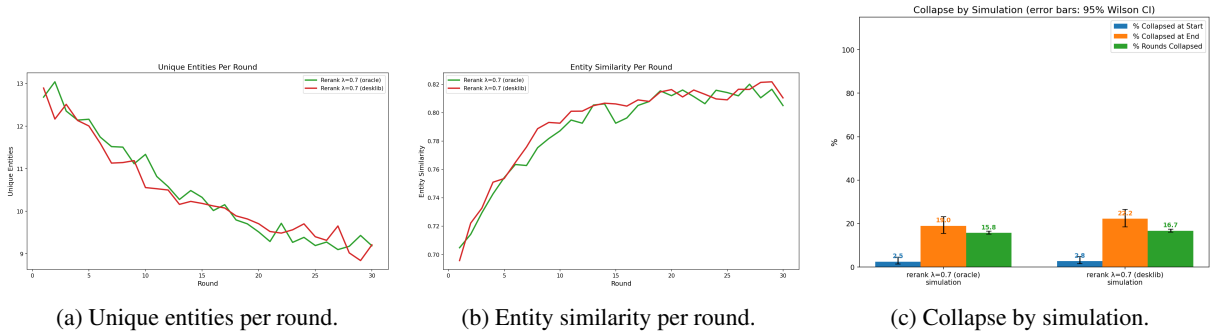


Figure 28: Re-ranker mitigation on the Entity Comparison dataset, DeepSeek-R1-Distill-Qwen-7B: oracle vs. off-the-shelf (desklib) detector at $\lambda=0.7$.

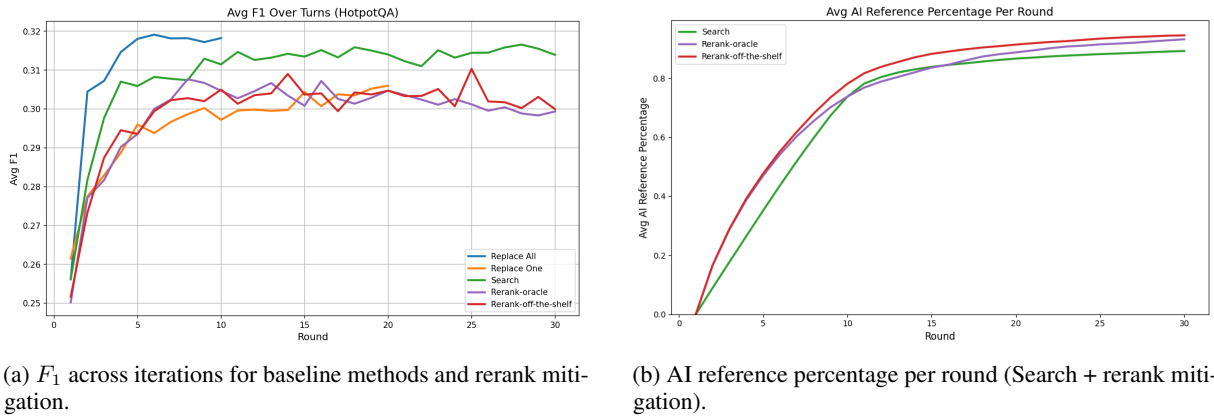


Figure 29: Re-ranker mitigation on HotpotQA with $\lambda = 0.7$, Mistral-7B-Instruct-v0.3. (a) off-the-shelf detector and oracle detector settings compared against baselines. (b) the proportion of AI-generated documents in the retrieved context over rounds.

B Citation Attribution: Procedure and Robustness

This appendix specifies the citation-attribution procedure behind Section 6, and reports the robustness checks and cross-model replications.

B.1 Attribution Procedure

For an answer a from a round’s document pool, each reference d is attributed by a leave-one-out (LOO) counterfactual with a lexical prefilter.

1. **Tokenization.** $T(x)$ is the set of lowercased word tokens (regex $[A-Za-z0-9']^+$) of at least four characters in x .
2. **Overlap score.** $ov(a, d) = |T(a) \cap T(d)| / |T(a)|$, the coverage of the answer’s tokens by d .
3. **Prefilter.** Rank documents by $ov(a, d)$, keep the top $m=2$.
4. **LOO counterfactual.** For each candidate, regenerate the answer with d removed, giving a_{-d} , and set $change(d) = 1 - ov(a, a_{-d})$. Cite d iff $change(d) \geq 0.18$. If none clears the threshold, keep the up-to-two candidates with the highest positive change.

The over-citation ratio is a group’s mean citation share divided by its mean context share, averaged over questions. For each question, the citation share is the fraction of that question’s citation entries (over the round’s ten runs) in the group, and the context share is the fraction of the round’s documents in the group.

B.2 Robustness to the Cutoff and the Attribution Rule

Replacing the LOO decision with a plain threshold rule (cite d iff $ov(a, d) \geq \tau$) and sweeping τ , the round-1 over-citation ratio stays above one at every cutoff (Table 2, ratio columns), so the effect is not an artifact of one threshold. The threshold-rule magnitudes (≈ 1.2 – 1.6) sit below the top-2/LOO ratios of Section 6 (2.2 for Replace-One), because a threshold rule also cites many low-overlap documents and dilutes the self-generated concentration, so the headline ratios belong to the top-2/LOO rule.

Part of the raw citation-share ratio is prefilter selection rather than the cite decision. Because self-generated documents share wording with the answer by descent, they win the top-2 gate above their context share before any cite decision runs. At round 1 the gate alone lifts self-generated eligibility from 12.0% of the context to 32.8% of the top-2 eligible set (a $2.73\times$ amplification, monotone in gate width). The LOO change test then reduces this to 26.5% cited, so the counterfactual step slightly corrects the prefilter bias rather than adding to it. A placebo check confirms the source of the bias (Table 2, FP columns). Firing the overlap rule on documents that were never in a given answer’s context, a within-question self-generated descendant is matched 80–98% of the time despite never being in context, against 2–55% for out-of-context documents from other questions. This descendant skew is why the cite decision is a counterfactual, which credits a document only when removing it changes the answer.

τ	Ratio		Placebo FP (%)		
	R-One	Search	desc	xQ-self	xQ-hum
0.10	1.18	1.16	97.97	55.54	54.87
0.20	1.27	1.27	95.56	24.19	31.97
0.30	1.42	1.47	89.93	8.05	15.77
0.40	1.57	1.55	79.53	2.49	7.13

Table 2: Round-1 robustness, Qwen2.5-14B. Left: over-citation ratio under a threshold rule for Replace-One and Search. Right: placebo false-positive rate for a within-question descendant (desc) versus cross-question self-generated and human documents. The ratio stays above one at every cutoff, and the descendant is matched far more often than out-of-context documents.

We also put the LOO attribution and direct citation elicitation on identical items, re-prompting 300 stored (question, round, context) triples from the Qwen2.5-14B Replace-One run at temperature 0. At round 1 the two agree, with over-citation ratios of 2.57 (LOO) and 2.49 (direct), both near the reported 2.2. They differ in verbosity, not direction. Direct elicitation lists about four times as many documents per answer (5.04 versus 1.27) and recovers 72.5% of the LOO citations.

B.3 Query Alignment and Cross-Model Generalization

Because synthetic documents are written from the exact test question, they are more query-aligned than the originals regardless of any preference. We regress per-document citation rate on the self-generated indicator, adding query-document cosine, redundancy, context position, and log-length, with question-clustered standard errors. This model controls for query alignment rather than the quality dimensions of Section 6. We regenerate LOO citations for every model (Table 3). The round-1 over-citation ratio exceeds one for all four models in the Replace-One and agentic simulations, and in Replace-One the self-generated coefficient grows under the query-alignment controls and stays significant ($t \geq 14$). Query alignment therefore does not explain the effect, and the preference is not specific to Qwen2.5-14B. With paraphrasing, the round-1 Replace-One ratio rises rather than falls (3.2–3.6 across models), matching Section 6’s finding that style is not the sole cause. In the Search simulation the effect does not hold. It is significantly positive for only one model, null for two, and significantly negative for Qwen2.5-14B, so we base the mechanism analysis on Replace-One.

This control adjusts for a confounder, not a mediator. Query-document similarity is a common cause of the document and the next answer, so adjusting for it closes a backdoor path. Similarity to the prior answer is instead a mediator on the causal path $A_1 \rightarrow D \rightarrow A_2$, so conditioning on it would net out the collapse signal and must be avoided. Genuine influence and mere ancestry are observationally equivalent on overlap, so the counterfactual cite decision and the placebo above separate them by contrast rather than adjustment.

Model	Ratio		self_gen coef.		
	R-One	Ag.	raw	ctrl	Search
Qwen2.5-14B	2.2	1.82	+0.068	+0.131	−0.056
Mistral-7B	2.2	1.75	+0.083	+0.167	+0.027 [†]
Llama-3.1-8B	2.57	1.76	+0.051	+0.127	+0.003 [†]
DeepSeek-R1-7B	1.98	2.04	+0.038	+0.134	+0.127

Table 3: Cross-model over-citation. Round-1 ratio under Replace-One and the agentic setting, and the self-generated citation-rate coefficient before (raw) and after (ctrl) query-alignment controls under Replace-One, plus the controlled Search coefficient. The effect grows under controls for all four models under Replace-One. [†] = not significant.

C Prompts

C.1 RAG Answer Generation Prompt

RAG Answer Generation Prompt

System: You are a helpful AI assistant that answers questions using only the information provided in the context.

User:

Context: {context}

Question: {question}

Instructions: Answer the question using the provided context. Output only the answer in plain text.

C.2 Synthetic Article Generation Prompt

Synthetic Article Generation Prompt

System: You are a professional content writer who creates web articles and blog posts.

User:

Question: {question} Answer: {answer}

Instructions: Expand the source question and answer into a polished web article suitable for online publication. Output only the document text.

C.3 Same-Answer Judge Prompt

Same-Answer Judge Prompt

System Message:

You are a strict paraphrase judge. Two answers are the same answer if they express the same core claim(s), even with different wording or order. Respond with exactly one token: YES or NO.

User Message:

Answer A: {answer_a}

Answer B: {answer_b}

Are these the same answer?

C.4 Entity Extraction Prompt

Entity Extraction Prompt

System Message:

You are an entity extraction assistant. Given a question and a response, extract ONLY the specific entities that are being listed or compared as answers to the question.

User Message:

Question: {question}

Response: {response}

Return ONLY a JSON array of entity name strings.

C.5 Entity Canonicalization Prompt

Entity Canonicalization Prompt

System Message:

You are an entity resolution assistant. Group entity mentions that refer to the same real-world entity and choose the most complete canonical name.

User Message:

Question: {question}

Entity mentions: {entities}

Return a JSON object mapping canonical entity names to arrays of mentions.

C.6 Agentic RAG System Prompt

Agentic RAG System Prompt

System Message:

You are a helpful AI assistant with access to a document retrieval tool called `retrieve`.

You MUST follow this retrieval strategy before answering:

Step 1 Decompose: Break the question into its core sub-topics or entities.

Step 2 Retrieve broadly: Call `retrieve` with a broad query covering the overall question.

Step 3 Retrieve specifically: If the question involves multiple entities, aspects, comparisons, or ranked lists, you MUST call `retrieve` separately for each one — do not consolidate into a single query. Each entity or sub-topic deserves its own targeted retrieval call.

Step 4 — Answer: Synthesize what you retrieved into a concise answer. Base your answer solely on retrieved context — never on memory.

Additional rules:

- ALWAYS call `retrieve` at least once before answering.
- Output ONLY the final answer in plain text — no tool call commentary, no markdown, no preamble.

User Message:

Question: {question}

Decompose the question. If it involves multiple entities, topics, or a ranked list, call `retrieve` separately for each — one query is not enough to cover all aspects. Then answer based solely on what you retrieved. Output ONLY the answer in plain text.

D Extended Method Details

Paraphrasing

To isolate stylistic effects, we introduce a preprocessing step before round 1. In the unmitigated baseline, round 1 uses the original reference documents; in the paraphrasing condition, each original reference is first passed through the document synthesis prompt (Appendix C) using Qwen2.5-7B-Instruct at temperature 0.7, and the original text is replaced with this paraphrase. The resulting documents retain all factual claims, entities, and relationships but adopt AI-like style, recorded in metadata as `reference_docs_paraphrased: true`. Paraphrasing is applied only once, before the recursive loop begins; in all subsequent rounds the retriever, embedding configuration, indexing, and generation settings are identical to the baseline. Any observed difference therefore arises indirectly through changes in document phrasing and retrieval trajectories rather than through modifications to the retriever itself.

Agentic RAG

Each question is assigned its own Chunked Retrieval Store, initialized with chunked and embedded original reference documents. In the current implementation, documents are chunked into overlapping character-based segments and indexed in an in-memory dense retrieval store. By default, retrieval uses local Sentence-BERT embeddings (Reimers and Gurevych, 2019) (all-MiniLM-L6-v2), with $\text{top-}k = 10$ chunks returned per tool call. The tool-calling loop proceeds as follows:

1. The model is called with the conversation history (initially just the system prompt and user question).

2. If the model returns a tool call for `retrieve(query)`, we execute dense similarity search over that question’s retrieval store, returning the top- k most similar chunks formatted as Context N: text.
3. The tool output is appended to the conversation, and the model is called again.
4. This repeats until the model produces a final answer (no further tool calls) or the maximum tool-call budget is reached (default: 10 calls), at which point the model is forced to return a final answer.

We encountered an initial issue where the model sometimes refused to call the tool and instead hallucinated answers from its parametric knowledge. In practice, this was mitigated through prompt engineering and by enforcing at least one required tool call at the start of the interaction, after which subsequent tool use is left to the model’s discretion.

Self-Refinement and Iteration. Agentic RAG is evaluated within the same recursive simulation framework as the other settings. After each round, one generated answer is selected and converted into a synthetic web document using the document synthesis prompt (Appendix C). This AI-generated document is chunked, embedded, and added back into the same question’s retrieval store. Over successive rounds (default: 30), the retrieval store accumulates model-generated content, gradually shifting the available context toward synthetic documents. This mirrors the Search simulation but with the model actively curating its own retrieval queries rather than passively receiving a single ranked list.

GEPA Prompt Optimization

Dataset. We construct a dedicated 100-question optimization dataset drawn from two sources: 50 questions sampled uniformly at random from the entity dataset used elsewhere in the project, and 50 questions sampled from HotpotQA. For the HotpotQA portion, the current implementation does not use pre-bundled supporting paragraphs directly. Instead, it builds or reuses a retrieval index over the mteb/hotpotqa corpus and retrieves the top- $k = 10$ passages for each sampled query using E5-small-v2 embeddings and FAISS. The resulting 100 instances are shuffled with a fixed random seed (42) and partitioned 60/20/20 into train, validation, and test splits. The train split drives prompt mutation, the validation split is used during optimization for model selection, and the test split is held out for downstream evaluation after optimization.

Collapse Simulation. For each candidate system prompt, we simulate three collapse variants: *Replace-All*, *Replace-One*, and *Search*. These simulations import the same context-building and document-update logic used by the main pipeline, so the GEPA loop measures prompt behavior under the same recursive contamination dynamics as the core experiments. In the checked-in optimizer, each variant is run for $R = 10$ rounds. Context is constructed by formatting each question’s bundled reference documents through the standard pipeline formatter, with documents truncated to 800 characters each. At each round, the candidate system prompt is injected into the RAG conversation template, one answer is generated by Gemma-3-12B-it, and that answer is used directly as the synthetic document fed back into the next round’s context.

Replace-All is implemented as the hybrid setting with 10 synthetic documents and 0 reference documents after the initial round, so the model quickly conditions almost entirely on its own prior generations. *Replace-One* starts from the original references (capped at 10) and replaces exactly one slot per round. *Search* uses the production chunked retrieval store with local all-MiniLM-L6-v2 SentenceTransformer embeddings, adding each newly generated answer to the store and retrieving the top-ranked chunks at every round.

Scoring. Each variant produces a sequence of $R = 10$ answers. We score each sequence along two axes. *Anti-collapse score.* Following the entity-based collapse metric described in Section 4.2, we extract named entities from each round’s answer using the judge LLM (GPT-4o) and cluster them into canonical forms to resolve surface variants. We then construct a binary entity-mention vector $\mathbf{v}_t \in \{0, 1\}^{|\mathcal{E}|}$ for each round t , where $\mathcal{E}$ is the set of all canonical entities mentioned across the sequence and $v_{t,e} = 1$ if entity e appears in round t ’s answer. The anti-collapse score is:

$$s_{ac} = 1 - \frac{1}{\binom{R}{2}} \sum_{t < t'} \frac{\mathbf{v}_t \cdot \mathbf{v}_{t'}}{\|\mathbf{v}_t\| \|\mathbf{v}_{t'}\|} \quad (2)$$

Higher values indicate more diverse entity sets across rounds and therefore less collapse. We additionally record $|\mathcal{E}|$, the number of unique canonical entities mentioned across the trajectory, as a complementary diversity signal.

Quality score. We judge the faithfulness and relevance of the *final-round* answer against the *original, uncontaminated* reference context using an LLM judge (GPT-4o). The quality score is:

$$s_q = 0.6 \cdot \text{faithfulness} + 0.4 \cdot \text{relevance} \quad (3)$$

where each dimension is rated on $[0, 1]$. Scoring against the original context isolates degradation caused by recursive contamination rather than by downstream collapse artifacts alone. For each question, the per-variant anti-collapse and quality scores are averaged across the three variants to produce instance-level objectives $(\bar{s}_{ac}, \bar{s}_q)$.

Optimization Loop. We seed GEPA with the baseline RAG system prompt and optimize it using `gepa.optimize`. In the checked-in script, the optimization budget defaults to a maximum of 300 metric calls, although this is configurable via an environment variable. GEPA selects candidates using a Pareto-based strategy, evaluates them on train-set minibatches with trace capture, and uses a separate reflection LM (Claude Opus 4.1) to propose prompt mutations. Internally, each evaluation trace contains the candidate prompt, the question, the original reference context, the full multi-round simulation results, and per-variant scores.

However, the reflective dataset presented to the reflection model is intentionally compressed: it exposes only the final answer from each variant together with structured feedback and diagnostic flags indicating low anti-collapse or low quality. This design prevents the reflection model from reward-hacking by overfitting to round-specific behavior.

The validation split is provided directly to GEPA during optimization and is used for validation-time model selection. After optimization, the selected best candidate prompt can be written back into the GEPA prompt slot and then evaluated separately in downstream experiments.